

On The Development of Knowledge Trees

Onri Jay Benally

September 2026

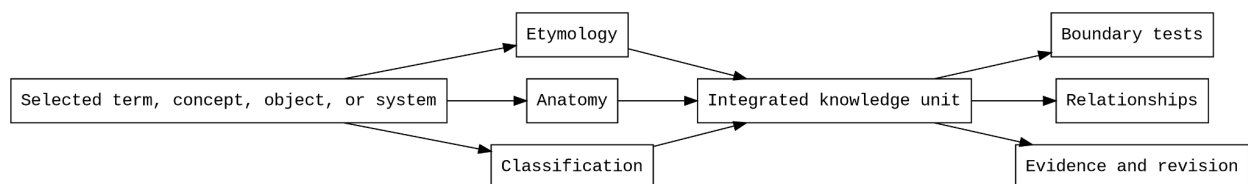
Abstract

Limited observation leaves gaps in knowledge, motivating systematic exploration of situations beyond direct experience. Within a recursive knowledge tree, combinations of root terms and adjacent concepts form candidate states whose properties change under defined transformations. Applied to assumption strength and thinking calibration, this approach yields six conceptual states, expanding to eighteen with the addition of three consequence categories. Comparing these candidates with recorded experience supports assessment of their applicability, providing a basis for further exploration. Across repeated cycles, pattern consolidation links recurring relationships to their contexts, providing a proposed pathway for intuition development. Feedback from observed outcomes permits reassessment of accumulated intuition, guiding subsequent inquiry without treating familiarity as confirmation. The resulting model organizes possibilities beyond personal experience, with predictive accuracy requiring empirical assessment.

Permutation and combination of root terms and related adjacent terms are similar to an interpolation process that can be produced on a table or spreadsheet. One can produce data (usually through computation) from a knowledge system. It is also possible to produce data by thinking of a model and creating a transformation process for that model. Data is generally valuable and is often used as a form of currency, so one should keep this in mind.

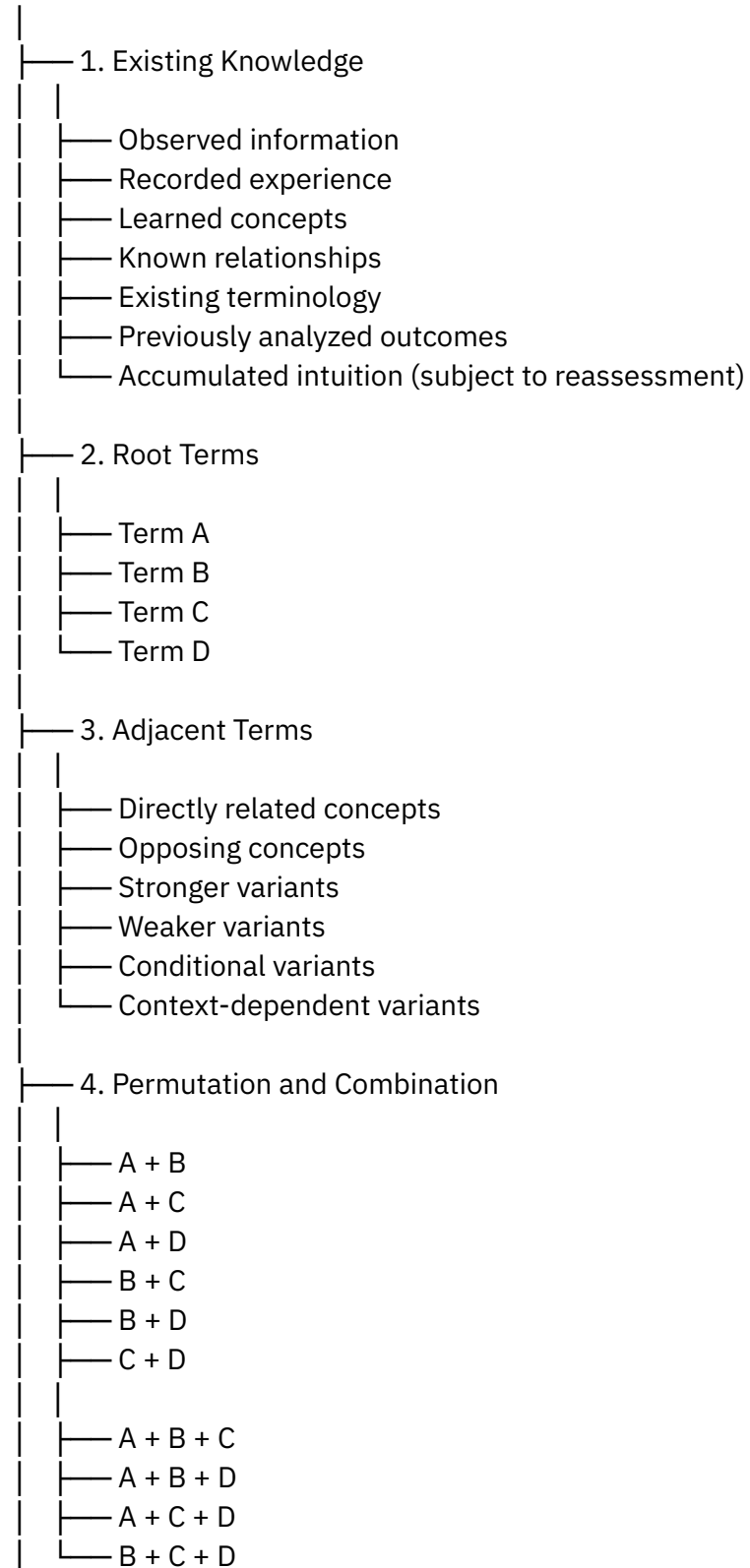
Knowledge trees can help visual interpolation and even extrapolation. One example I came up with is related to the term "assumptions", which branched off into "weak assumptions" and "strong assumptions", which branched off into more branches like "weak assumptions + overthinking" and 5 more branches for a total of 6 branches, which could lead to even more branches. By doing so, we can achieve the ability to "know" or highly anticipate the future without actually having to experience it for ourselves, since "certainty" in itself is simply a human concept. Things happen in the so-called universe all the time regardless of whether humans think something should happen or not or are completely uncertain about due to the limited existence of humanity itself in the timeline applied to the "universe" or lack of awareness about that which is not classified.

A general-purpose knowledge-generation system can organize existing concepts into a tree of candidate states. Combining related concepts and applying transformations produces states for analysis and further expansion. Within this conceptual model, interpolation describes exploration within a known region, and extrapolation describes extension beyond that region.



1. Overall knowledge-generation tree

Knowledge Generation



5. Transformation

- Combine concepts
- Modify assumptions
- Change context
- Add constraints
- Remove constraints
- Reverse relationships
- Increase intensity
- Decrease intensity

6. Generated Knowledge States

- Previously observed state
- Closely related state
- Plausible unobserved state
- Edge-case state
- Contradictory state
- Novel composite state

7. Analysis

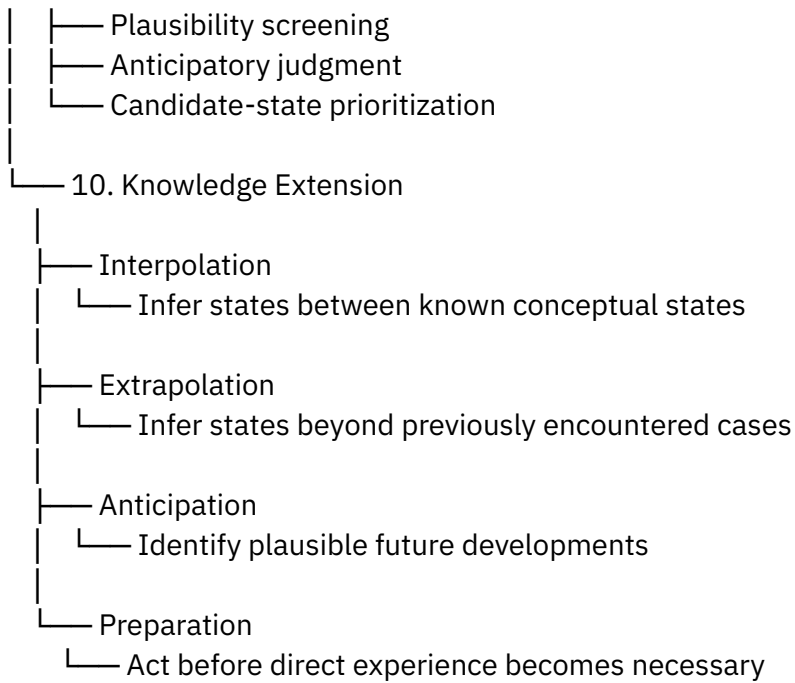
- Internal consistency
- Similarity to known cases
- Causal plausibility
- Context compatibility
- Frequency/ recurrence
- Consequence severity
- Information value

8. Pattern Consolidation

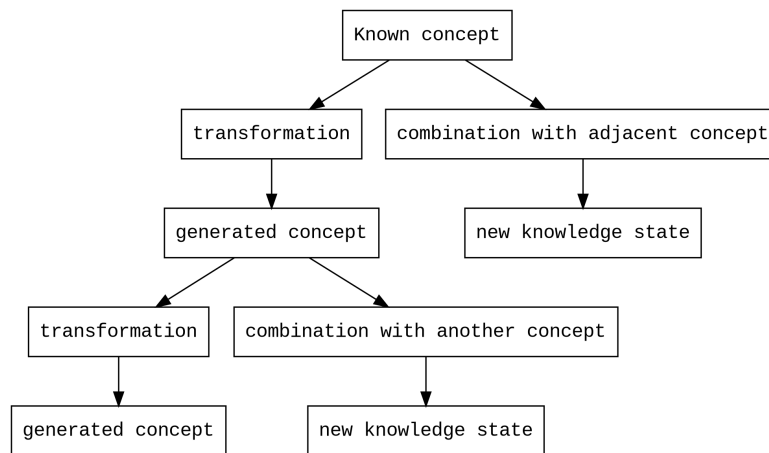
- Compare recurring relationships
- Associate patterns with context
- Compare predictions with observed outcomes
- Correct errors across cycles

9. Intuition Development

- Rapid pattern recognition
- Pattern completion
- Anomaly detection



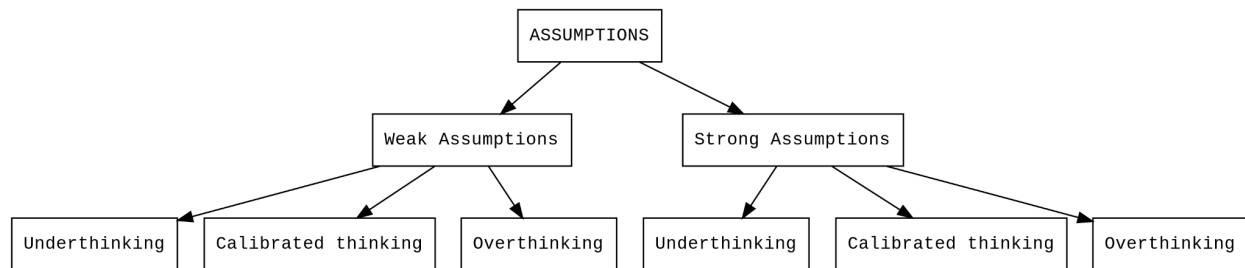
Generated combinations form new nodes that can serve as roots for subsequent expansion.



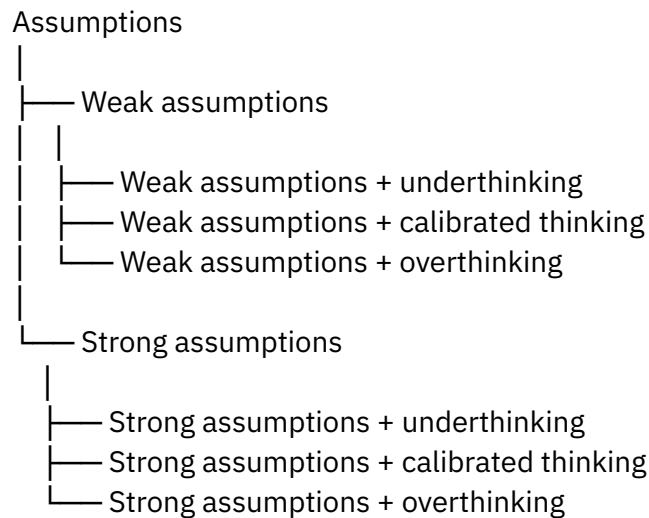
Repeated application of this process produces a recursively expanding knowledge tree.

2. Assumption strength and thinking calibration

The initial generation uses assumption strength and thinking calibration as two conceptual axes.



A conventional ASCII tree represents the same branching structure below.



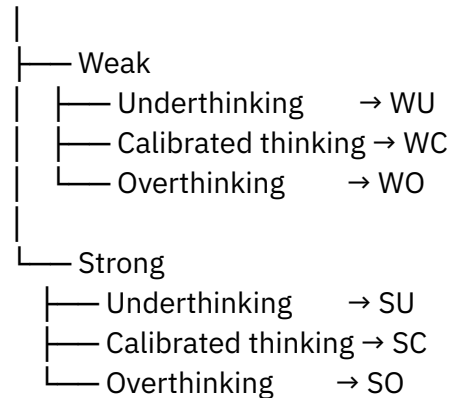
A tabular representation assigns one row to each combination, allowing the same conceptual space to be organized in a spreadsheet.

Assumption strength	Cognitive processing	Consequence	Generated state
Weak	Underthinking	Low	W-U-L
Weak	Underthinking	Medium	W-U-M
Weak	Underthinking	High	W-U-H
Weak	Calibrated thinking	Low	W-C-L
Weak	Calibrated thinking	Medium	W-C-M

Assumption strength	Cognitive processing	Consequence	Generated state
Weak	Calibrated thinking	High	W-C-H
Weak	Overthinking	Low	W-O-L
Weak	Overthinking	Medium	W-O-M
Weak	Overthinking	High	W-O-H
Strong	Underthinking	Low	S-U-L
Strong	Underthinking	Medium	S-U-M
Strong	Underthinking	High	S-U-H
Strong	Calibrated thinking	Low	S-C-L
Strong	Calibrated thinking	Medium	S-C-M
Strong	Calibrated thinking	High	S-C-H
Strong	Overthinking	Low	S-O-L
Strong	Overthinking	Medium	S-O-M
Strong	Overthinking	High	S-O-H

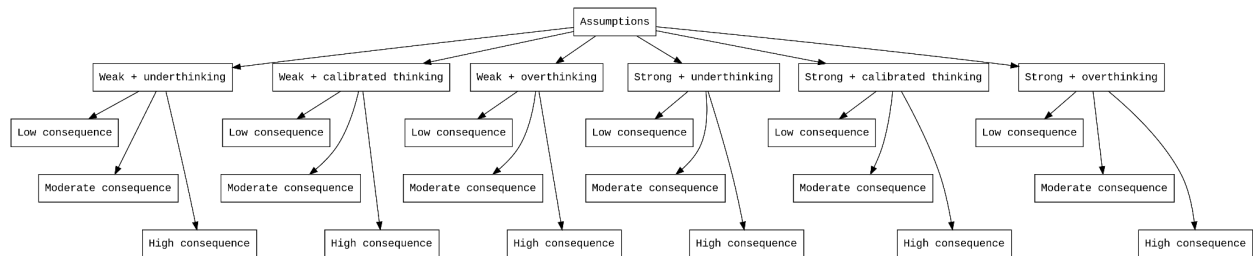
Combining the two levels of assumption strength with the three categories of cognitive processing yields six states.

Assumption Strength



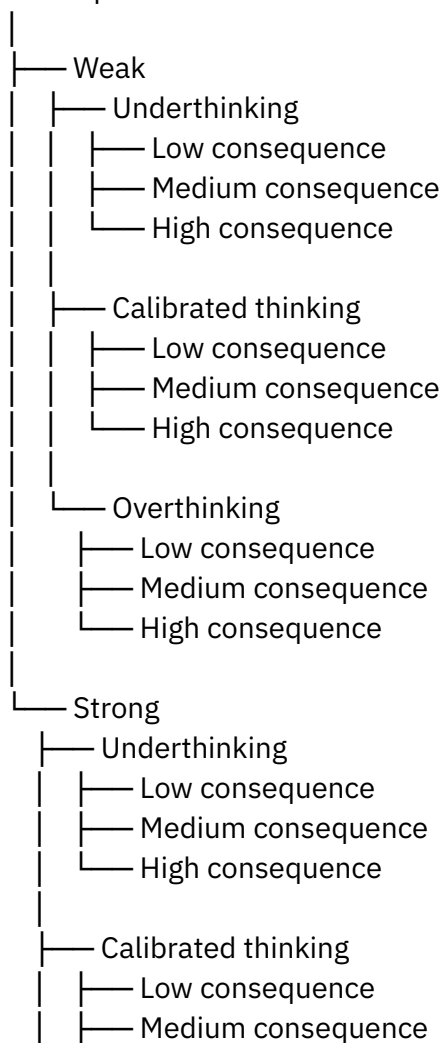
In this model, **calibrated thinking** denotes an amount and depth of thought proportionate to the demands of a situation. It occupies the conceptual region between insufficient and excessive cognitive processing.

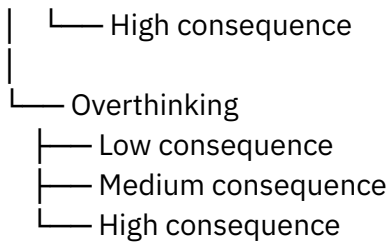
Each of the six nodes can branch further according to consequence severity.



Including three consequence categories expands the six-state structure to 18 combinations, as shown in the equivalent hierarchical arrangement below.

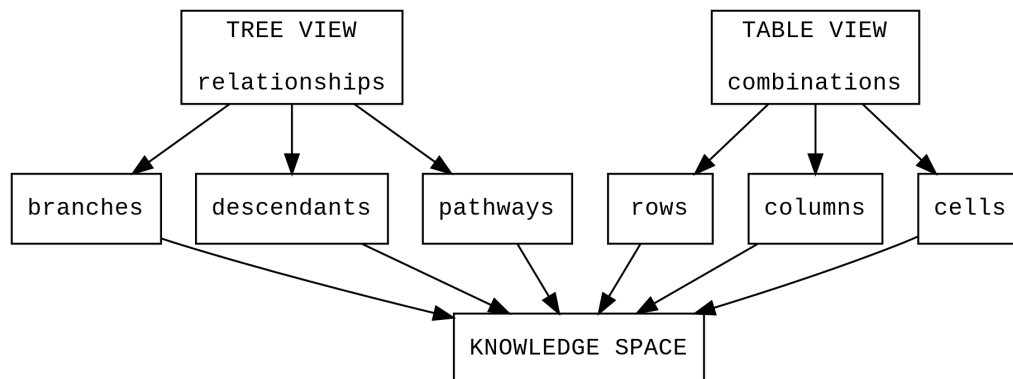
Assumptions





With three conceptual axes, this version produces $2 \times 3 \times 3 = 18$ distinct conceptual states.

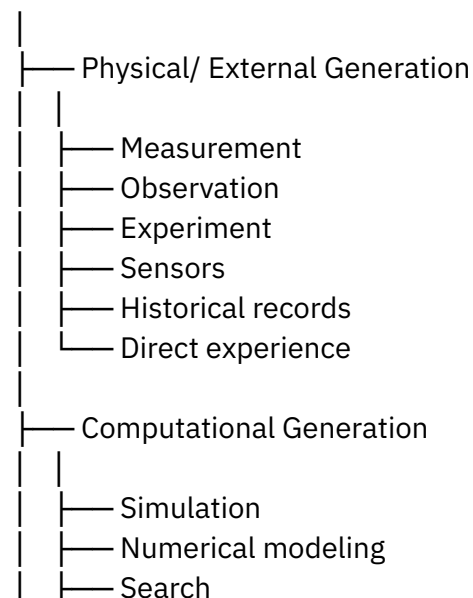
The tree emphasizes hierarchical relationships, and the table enumerates the corresponding combinations within the same conceptual space.

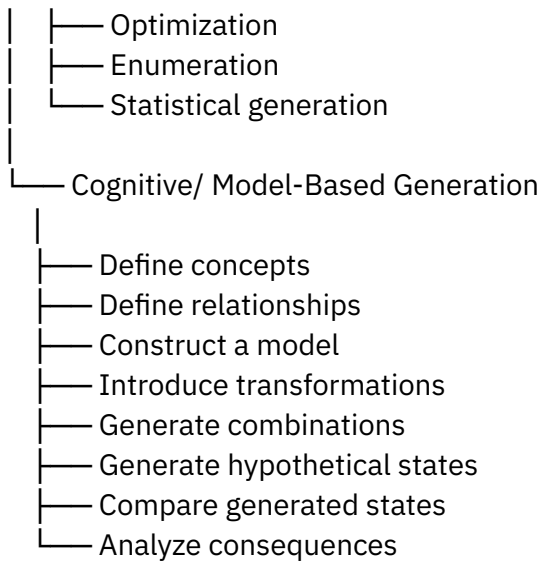


3. Cognitive modeling as a data-generation mechanism

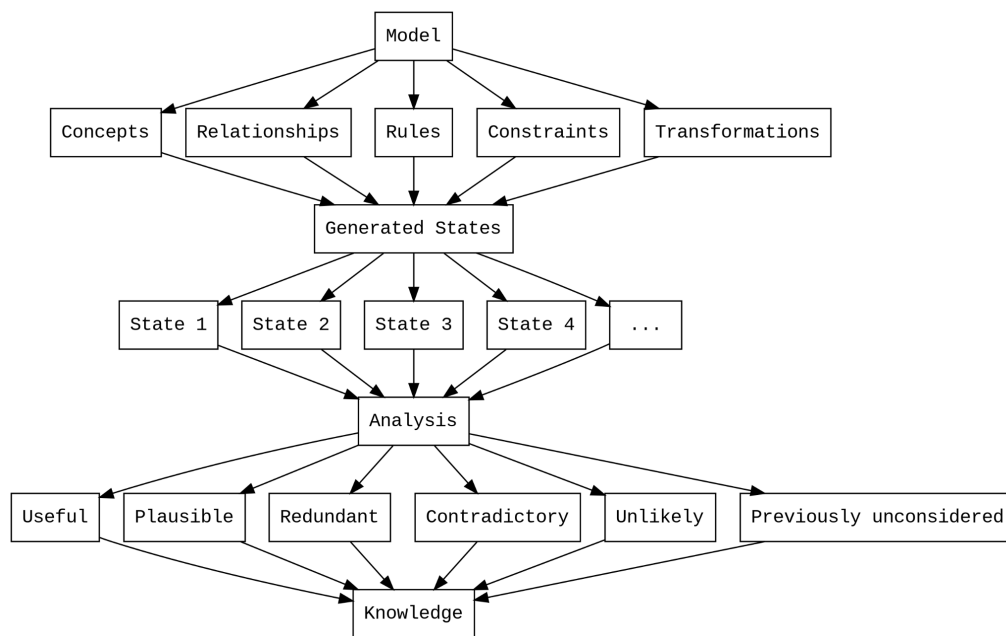
Data generation can be organized by the processes that produce observations, computational outputs, and conceptual candidates.

Data Generation





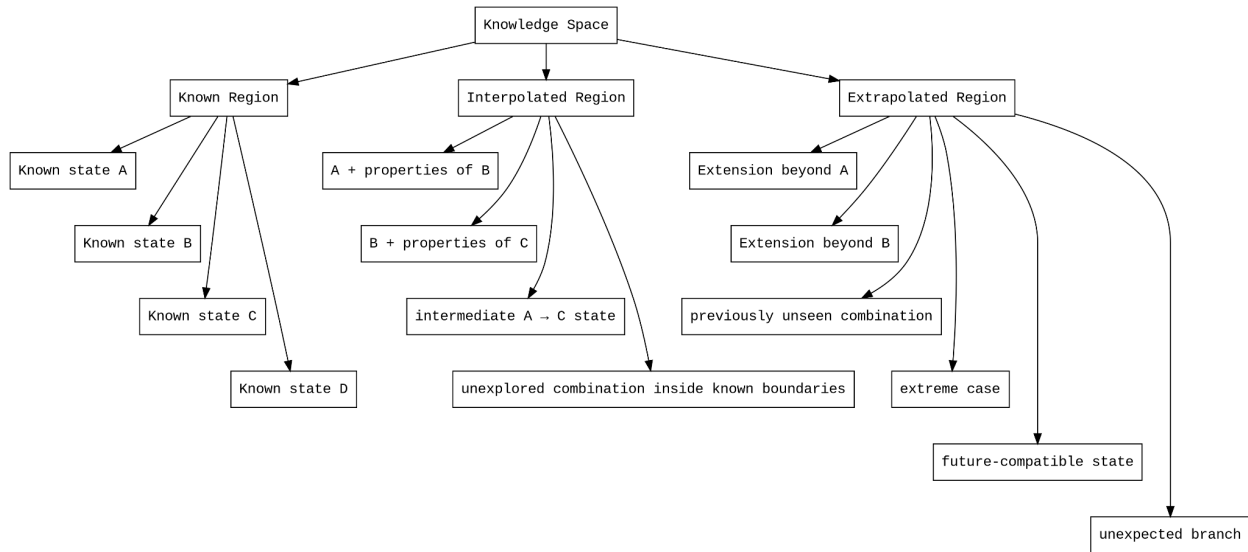
Within the model-based pathway, transformations generate candidate states for subsequent analysis.



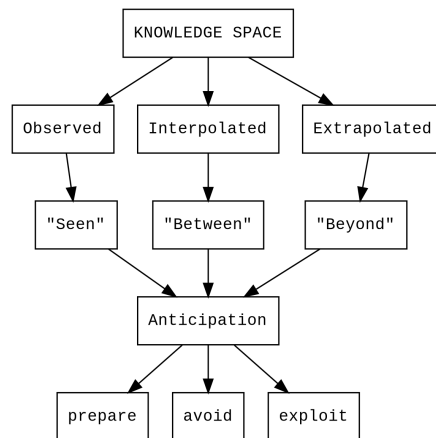
A defined transformation system can generate structured candidate data in addition to data obtained from measurement. Analysis evaluates the informational value of each candidate; empirical support requires comparison with observations.

4. Interpolation and extrapolation in a general knowledge tree

Interpolation and extrapolation serve here as conceptual analogies for exploration within and beyond known regions.

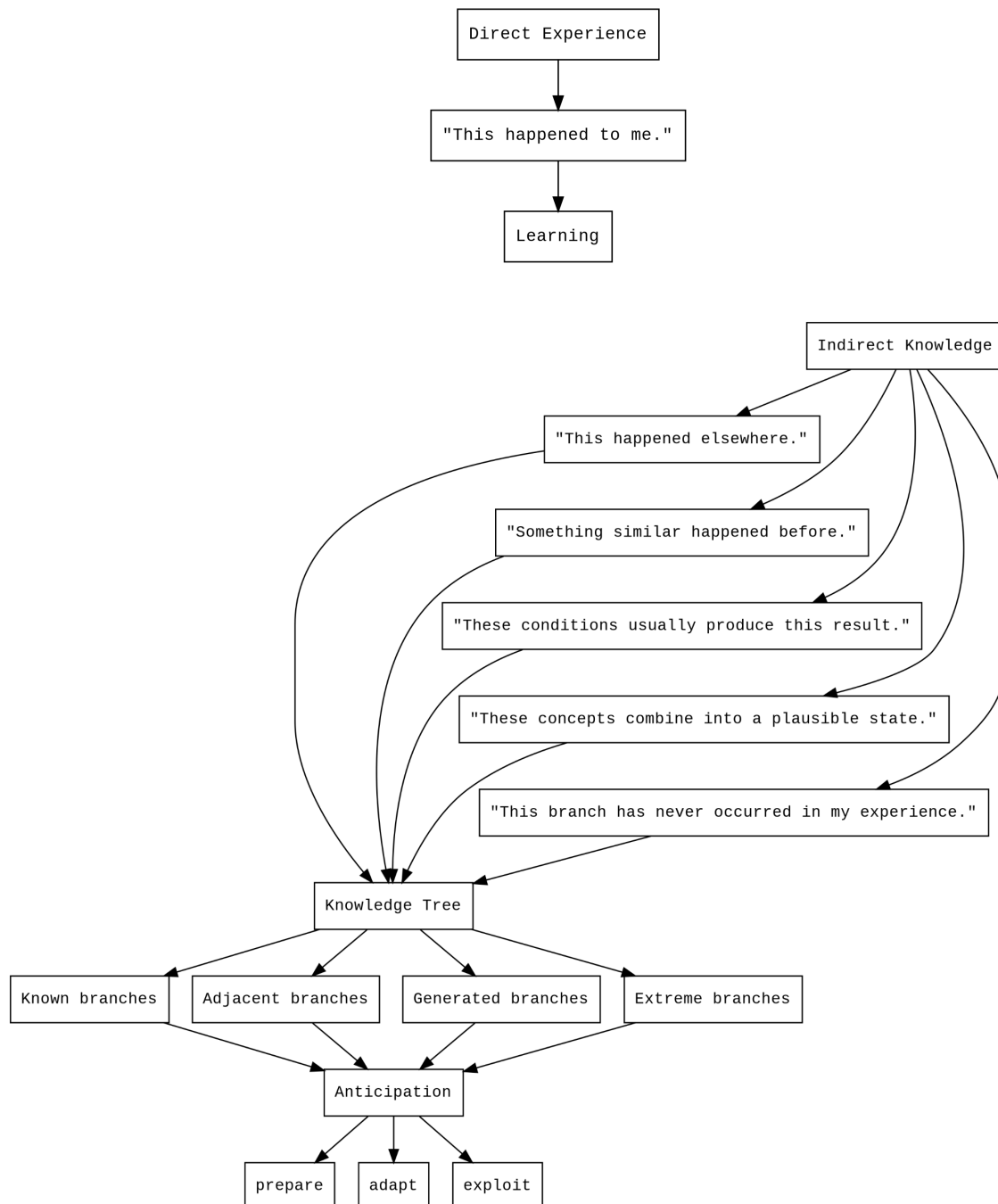


The following representation connects observed, interpolated, and extrapolated states to anticipation and possible responses, including preparation, avoidance, and exploitation.

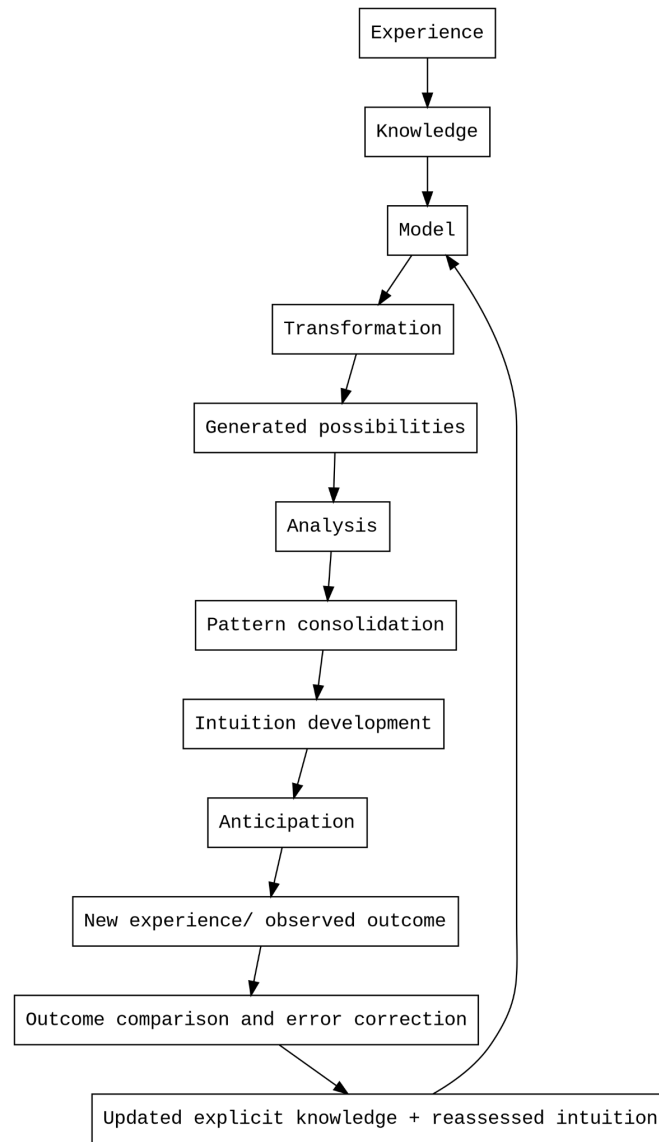


5. Anticipating events without personally encountering them

Indirect knowledge and generated scenarios can support anticipation of situations beyond personal experience.



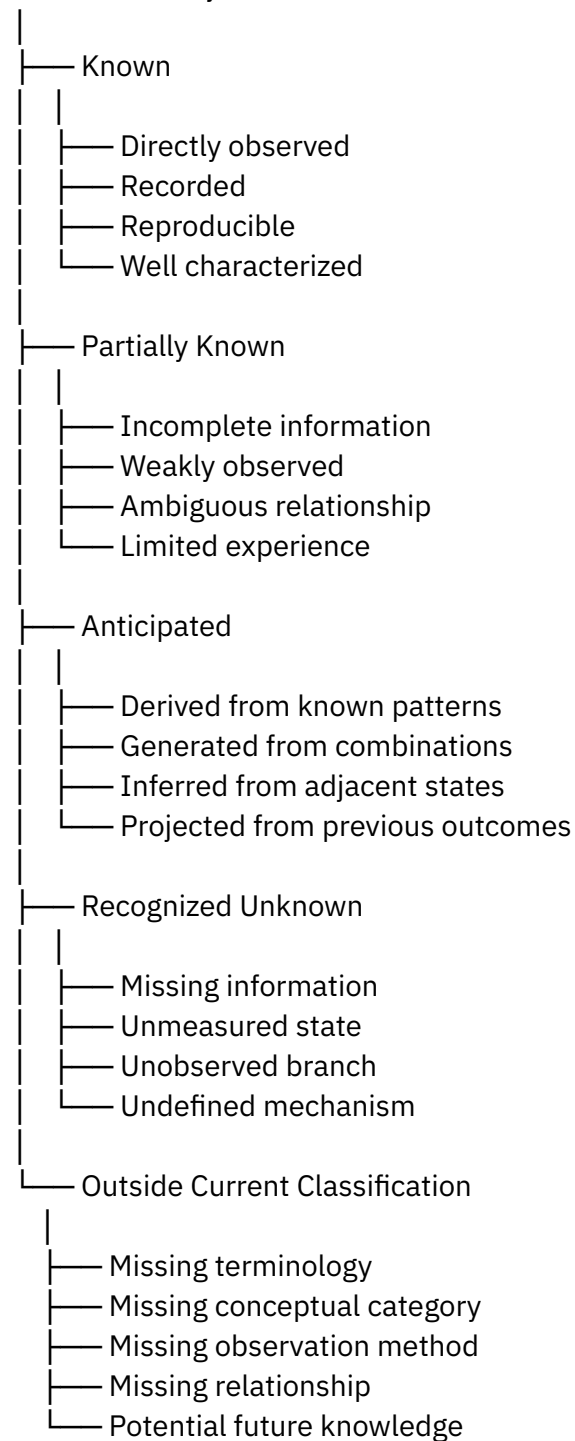
A recursive representation incorporates new experience into the model after each cycle of analysis and anticipation.



6. Unknowns can themselves branch

The classification can extend to incomplete information, recognized gaps, and phenomena outside the available conceptual categories.

Possible Reality

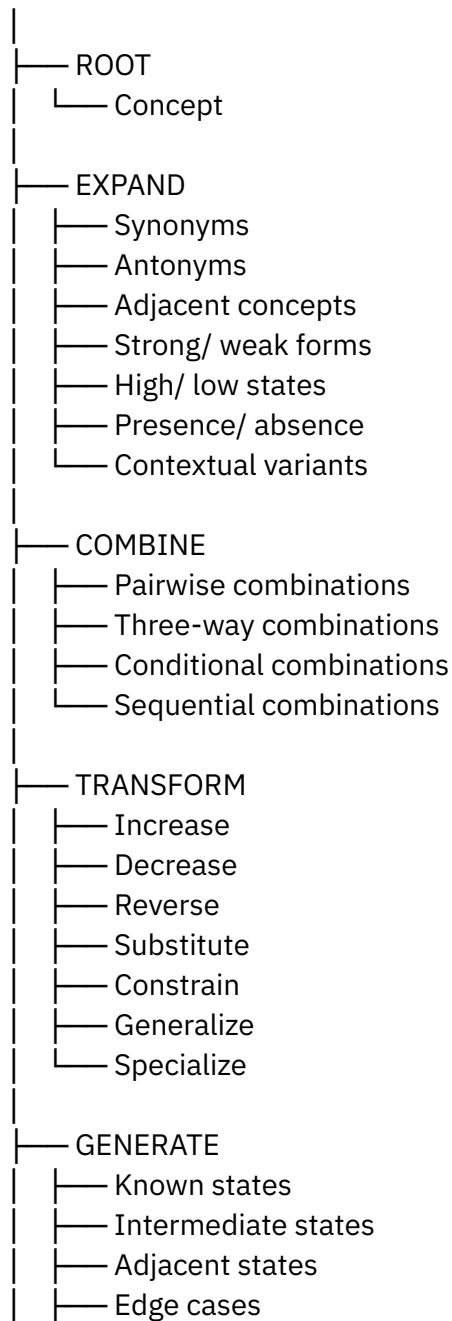


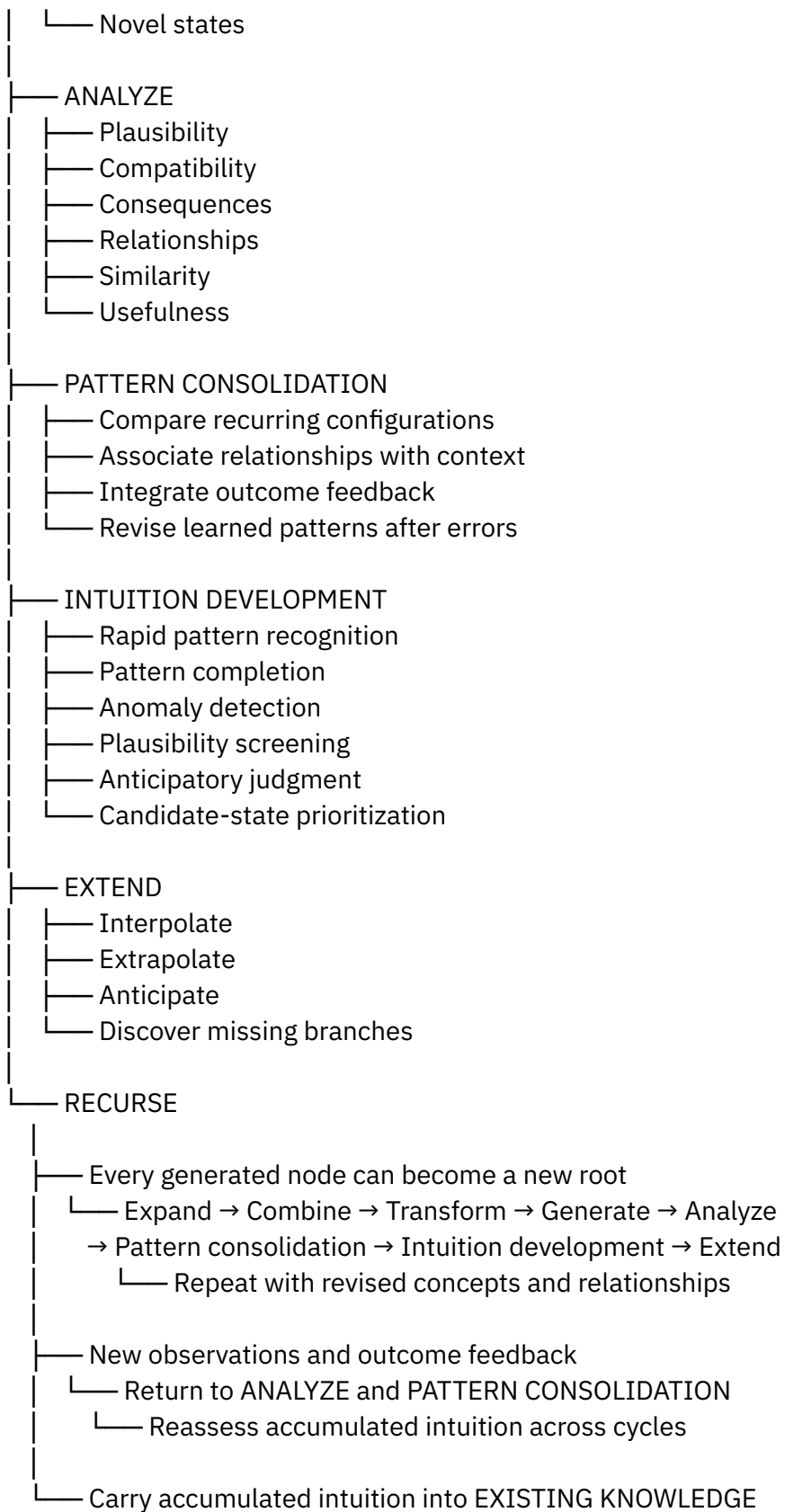
The final branch represents limits imposed by the vocabulary and classifications used to construct a knowledge system. Phenomena may occur beyond the descriptive capacity of the current tree, motivating the addition of terms, categories, or observation methods.

7. The recursive "knowledge tree engine"

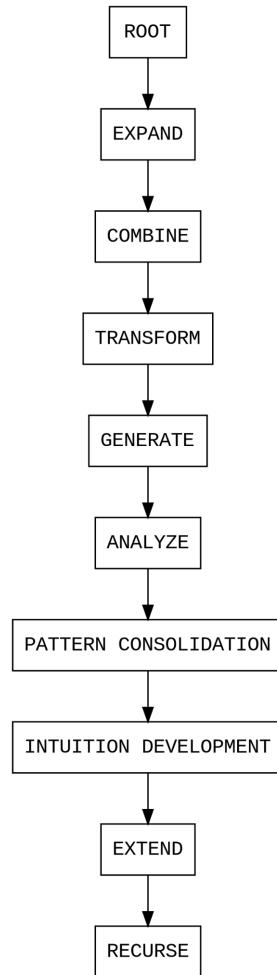
The proposed architecture organizes conceptual expansion, combination, transformation, and analysis into a recursive process.

KNOWLEDGE TREE ENGINE





The following sequence summarizes the recursive process.



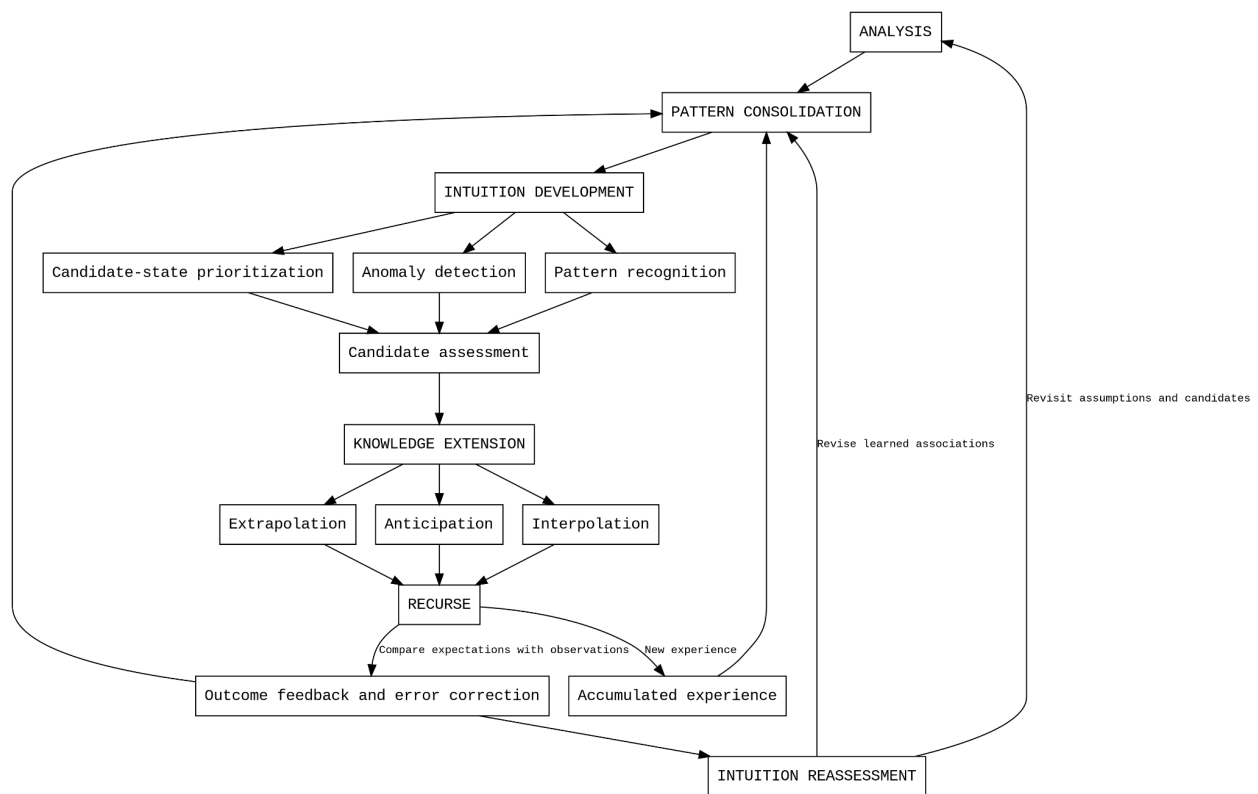
Starting from a concept such as **assumptions**, the process constructs a structured space of adjacent, intermediate, extreme, and previously unconsidered states. Trees, adjacency graphs, matrices, and spreadsheets provide alternative representations of this space for further expansion. Generated states retain the status of candidates until their applicability has been assessed.

8. Intuition development as a convergence and feedback process

Within the proposed knowledge-tree model, intuition development describes the gradual development of learned pattern recognition across repeated cycles of exploration, comparison, and outcome feedback. It occupies the transition between analysis and extension, where recurring relationships can guide the selection of possibilities for further examination. Developed intuition concerns how familiar configurations are recognized; the thinking-calibration categories in Section 2 concern the amount and depth of deliberate processing.

Within each cycle, **intuition development** describes a process of accumulation, refinement, contextualization, and correction. **Accumulated intuition** denotes the learned pattern knowledge carried into a later cycle as part of existing knowledge. Reassessment can then strengthen a useful association, narrow its scope, or weaken an earlier expectation after contradictory outcomes.

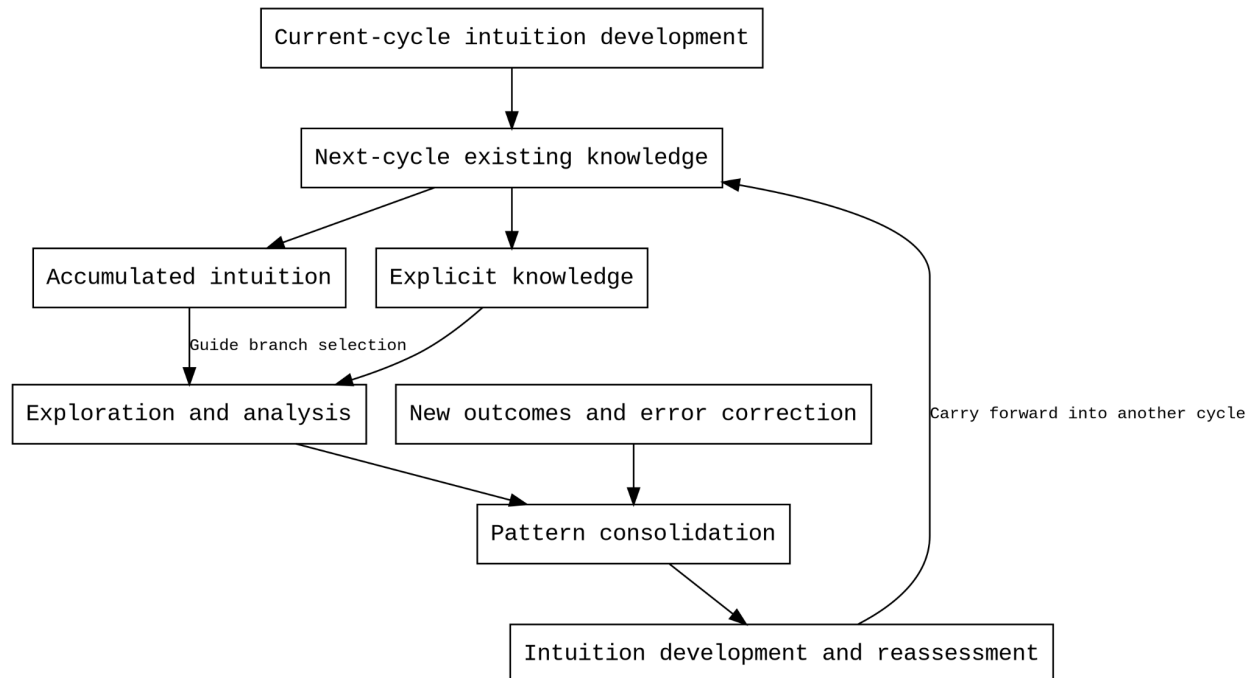
The following convergence places pattern consolidation at the meeting point of analysis, accumulated experience, and outcome feedback. Intuition development follows this convergence and guides the selection of candidate states for further assessment.



The return paths connect observed outcomes with both explicit analysis and learned associations. Repeating a generated scenario can increase familiarity without establishing that its anticipated consequences occur in practice. Intuition guides recognition and prioritization; assessment determines whether a proposed extension is supported or not.

Term	Function
Pattern consolidation	Recurring relationships acquire recognizable structure as cases and contexts are compared.
Intuition development	Learned structures become available for rapid recognition and prioritization, with refinement across cycles.
Accumulated intuition	Developed pattern knowledge is carried into subsequent cycles as part of existing knowledge.
Intuition reassessment	New outcomes prompt revision, contextual restriction, or weakening of earlier pattern associations.
Knowledge extension	Assessed candidates and developed understanding support interpolation, extrapolation, and anticipation.

Distinguishing between process and carried state gives intuition two complementary roles. Analysis and feedback develop intuition during the current cycle; in the next, this accumulated intuition guides branch selection and aids assessment, alongside explicit knowledge.



Component	Contribution to intuition development	Role in subsequent exploration
Pattern exposure	Encounters provide configurations that can be compared across cases.	Familiar features help identify relevant branches.
Pattern comparison	Similarities and differences reveal which relationships recur.	Candidate states can be grouped or separated by their structure.
Outcome feedback	Observed consequences can be compared with earlier expectations.	Successful and failed predictions inform later prioritization.
Error correction	Mismatches prompt changes to assumptions or contextual boundaries.	Later cycles can revise or discard misleading patterns.
Relationship compression	Several explicit relationships may be represented as a familiar configuration.	Recognition can suggest a likely consequence with less conscious enumeration.
Context association	Learned patterns are linked to the circumstances in which they were encountered.	Applicability can be reassessed as conditions change.
Recursive cycling	Further exploration revisits patterns alongside new information.	Accumulated intuition can be refined and carried into existing knowledge.

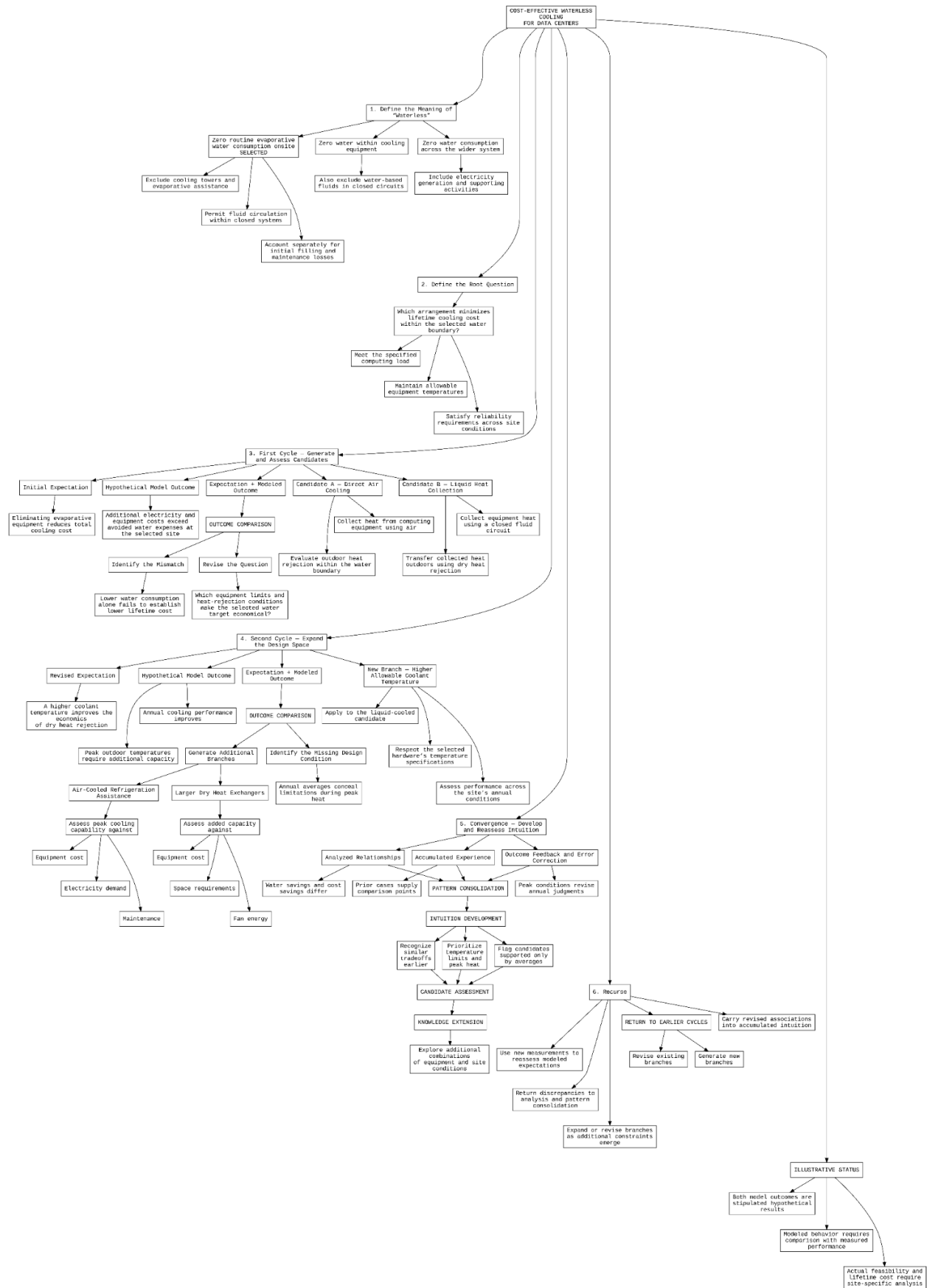
This modeled way of thinking pairs an explicit reasoning pathway with a learned-recognition pathway. Explicit reasoning generates concepts, combinations, transformations, and candidate assessments. Once repeated comparison and corrective feedback make familiar relationships more accessible, learned recognition can then guide further explicit exploration. Consequently, a knowledge tree can facilitate intuition development. While explicit exploration supplies relationships for comparison, feedback corrects accumulated pattern knowledge before it informs another cycle of anticipation.

Illustrative feedback example: Data Center Cooling

Cooling a data center without routine evaporative water consumption presents a design problem in which lower water demand may conflict with lifetime cost. In this hypothetical example, comparing expected savings with modeled performance exposes a limitation in selecting equipment solely by its water consumption. Further exploration introduces the allowable coolant temperature as a design dimension, illustrating how feedback revises initial assumptions and redirects the search toward candidates suited to the facility's climate.

Stage	Illustrative development	Contribution to the knowledge tree
Root question	Which cooling arrangement meets the specified computing load and temperature limits at the lowest lifetime cost, without routine evaporative water consumption?	Defines a problem with assessable constraints.
Initial branches	Compare direct air cooling with liquid heat collection connected to dry heat rejection.	Separates heat collection inside the facility from heat rejection outdoors.
First expectation	Eliminating evaporative equipment will reduce total cooling costs.	Makes an assumption available for assessment.
Hypothetical first outcome	For the selected site, a model predicts that additional electricity and equipment costs exceed the avoided water expenses.	Revises the assumption that water savings necessarily produce financial savings.
New branch	Investigate a higher allowable coolant temperature within the selected hardware's specifications.	Expands the search from equipment substitution to compatibility between computing hardware and cooling conditions.
Hypothetical second outcome	Annual performance improves, but peak outdoor temperatures still require additional cooling capacity.	Reveals a limitation hidden by annual averages.

Stage	Illustrative development	Contribution to the knowledge tree
Further expansion	Compare larger dry heat exchangers with air-cooled refrigeration assistance during peak conditions.	Introduces alternative responses to the newly identified constraint.
Intuition reassessment	Future evaluations examine peak conditions and temperature limits earlier in the search.	Shows how corrected expectations guide subsequent reasoning.



Conclusion

Recursive knowledge trees organize conceptual possibilities into an expandable structure, making relationships available for comparison across successive cycles. Within this structure, intuition development connects recurring patterns with the selection of candidates for further assessment. Accumulated intuition carries learned associations into subsequent exploration, subject to correction as observed outcomes challenge earlier expectations. The eighteen-state example demonstrates enumeration of all provided eighteen combinations within selected categories, with broader coverage requiring additional concepts or revised assumptions. Predictive accuracy and improvements in intuitive judgment require empirical testing beyond the enumeration of possibilities. As a conceptual model, the approach supports structured anticipation grounded in assessed relationships, preserving the capacity to revise expectations as experience expands.